

# Quantizing ESM-2 for Protein Language Modelling: A Measurement Study on Speed, Memory, and Accuracy

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*Benchmarking report — ESM2-3B on NVIDIA H200 and A100*

7 August 2026

## Abstract

We benchmark six numerical precision configurations for ESM-2 protein language models across three axes — throughput, memory footprint, and predictive accuracy — on two workloads with sharply different characteristics: bulk embedding extraction and deep mutational scanning (DMS) variant-effect scoring. Accuracy is evaluated on the complete ProteinGym substitution benchmark — 201 assays, 2.41M variants — at **three model scales spanning 650M to 15B parameters**, with a paired bootstrap clustered on protein. Three findings follow. First, **benchmark averages conceal the failure that decides deployability**: no configuration shifts mean correlation by more than 0.007 at any scale, yet INT8 dynamic quantization — indistinguishable from fp32 on that mean at 3B ( $p = 0.34$ ) — takes a single assay from  $\rho = 0.591$  to 0.223. Second, **fidelity against fp32 bounds risk but cannot rank quality**: over 3015 assay/configuration pairs it predicts the *magnitude* of ground-truth change ( $r = 0.56$ – $0.81$ ) but not its *direction*, and INT4 weight-only quantization is significantly harmful at 650M, significantly *beneficial* at 3B, and null at 15B — a reversal with no monotone trend to extrapolate. Third, **the configuration ranking is itself scale-dependent**: FP8 dynamic quantization runs at  $0.30\times$  bf16 DMS speed at 650M and  $0.97\times$  at 15B while halving memory, so the workload conflict that dominates at small scale dissolves at large scale. We also report four measurement artifacts encountered during this study, three of which inverted the result they were meant to measure.

## 1 Introduction

Quantization is usually presented as a straightforward trade: reduce numerical precision, gain speed and memory, lose a little accuracy. This framing is inherited from large autoregressive language models, where it is broadly correct. We set out to test whether it transfers to ESM-2, a family of protein language models, and found that essentially every part of it requires qualification.

ESM-2 is a BERT-style bidirectional *encoder*. It performs a single forward pass with no KV cache and no autoregressive decode. This places it in a **compute-bound** regime rather than the memory-bandwidth-bound regime that governs LLM inference at batch size 1. The distinction is not academic: the dominant open-source quantization toolchains (GPTQ, AWQ, bitsandbytes NF4, GGUF) are all *weight-only*. They store INT4/INT8 weights and dequantize to bf16 immediately before an ordinary bf16 matrix multiply. When the bottleneck is memory bandwidth, this is a large win. When the bottleneck is arithmetic, as it is here, it adds work to an already-saturated kernel.

This report addresses three questions:

1. What do the available quantization configurations actually cost and deliver on the three axes of speed, memory, and accuracy?
2. Does the answer depend on the workload? (It does, and the two workloads we study prefer opposite configurations.)
3. Do the fidelity metrics conventionally used to validate quantization predict real-world predictive accuracy? (Only partially, and in a way that matters.)

## 2 Experimental Design

### 2.1 Configurations

Six configurations were evaluated, all applied via `torchao` to the encoder `Linear` layers only:

Table 1: Quantization configurations. Weight-only variants target memory; the dynamic (W8A8) variants also quantize activations and can therefore use low-precision tensor cores for the matrix multiply itself.

| Configuration         | Class       | Description                                                                             |
|-----------------------|-------------|-----------------------------------------------------------------------------------------|
| <code>fp32</code>     | baseline    | Reference for all fidelity measurements.                                                |
| <code>bf16</code>     | baseline    | The practical production default.                                                       |
| <code>int8_wo</code>  | weight-only | INT8 weights, bf16 compute. Memory-oriented.                                            |
| <code>int4_wo</code>  | weight-only | INT4 weights, bf16 compute. Smallest footprint.                                         |
| <code>int8_dyn</code> | W8A8        | INT8 weights <i>and</i> dynamically quantized activations; uses INT8 tensor cores.      |
| <code>fp8_dyn</code>  | W8A8        | FP8 (E4M3) weights and activations. Requires compute capability $\geq 8.9$ ; H200 only. |

### 2.2 Workloads

**Bulk embedding extraction.** Large length-bucketed batches under a token budget, measured in residues per second. This is the compute-bound regime.

**DMS variant-effect scoring.** The standard ESM masked-marginals protocol: for each mutated position, mask it and record  $\log p(\text{mut}) - \log p(\text{wt})$  from the resulting distribution. Cost scales with the *number of mutated positions*, not with the number of variants, and the masked batch is narrow. This is the latency-bound regime.

Two optimizations were applied to the DMS path before benchmarking, both of which are absent from typical reference implementations: only positions carrying a variant are masked, and the masked copies are batched rather than run one at a time.

### 2.3 Metrics

Three fidelity metrics were recorded, deliberately ordered by sensitivity:

1. **Embedding cosine similarity** (mean-pooled, excluding special tokens and padding). Forgiving; error averages out along the sequence.
2. **Logit KL divergence**, per residue. Moderately sensitive.

3. **DMS Spearman  $\rho$  against fp32.** Most sensitive, because the score is a *difference of two near-equal log-probabilities*, so quantization noise does not cancel and is amplified relative to signal.

All three are drift-from-fp32 measures. They quantify how far a configuration moves the answer, not whether it moves toward or away from the truth. To measure the latter, we added a fourth metric requiring real experimental data: **Spearman  $\rho$  against wet-lab DMS measurements**, from ProteinGym.

## 2.4 Real assay data

Three ProteinGym substitution assays were selected to span the sequence-length range, since peak memory during masked-marginals scoring grows with  $L$ :

Table 2: ProteinGym assays used. All 19,557 single-substitution variants had their stated wild-type residue verified against the reference `target_seq` before use; all matched. This check is load-bearing — an off-by-one in position indexing does not raise an error, it silently scores the wrong positions and yields a plausible-looking correlation.

| Assay                    | $L$ | Singles | Masked positions | Coverage |
|--------------------------|-----|---------|------------------|----------|
| IF1_ECOLI_Kelsic_2016    | 72  | 1,367   | 72               | 100%     |
| BLAT_ECOLX_Stiffler_2015 | 286 | 4,996   | 263              | 92%      |
| HSP82_YEAST_Flynn_2019   | 709 | 13,194  | 707              | 100%     |

## 2.5 Full benchmark data

The three-assay analysis is superseded by a run over the complete ProteinGym substitution benchmark. All 217 assays were extracted from the `ProteinGym.v1` parquet release and every one passed wild-type validation across 2,465,767 variants. The 201 assays fitting ESM-2’s 1022-residue context were scored: 2,413,913 variants over 40,775 masked positions, spanning  $L = 37$  to 934 and five function categories.

The masked-position count, not the variant count, sets the cost: masked-marginals scoring performs one forward pass per mutated position, so 2.41M variants are nearly free on top of 40,775 forwards.

An earlier three-assay path combined per-assay CSVs from the `v0.1` release with wild-type sequences from the `v1.0` reference file. The two agree on the 85 assays present in both and diverge elsewhere, including renames. That is acceptable for three hand-picked assays and unsafe at benchmark scale; the `v1` parquet carries `target_seq` inline, so scores and wild-type sequences come from a single release.

## 2.6 Comparability to published benchmark numbers

Two choices make our absolute correlations not directly comparable to a published ESM-2 ProteinGym figure.

**The aggregation convention matters more than any effect we measure.** An unweighted mean over assays, a mean of per-protein means, and a mean of per-category means are all defensible and give different answers spanning 0.019 — roughly three times the largest quantization effect here.

**The 16 excluded assays are not a random sample.** They exceed ESM-2’s context ( $L = 1154-3423$ ) and are 2.1% of benchmark variants, but 50% are low-MSA-depth against 13.9% retained,

Table 3: fp32 mean Spearman under four aggregation conventions. The flat mean we report is the most favourable of the four. The *differences between configurations* are stable across all four: signs identical, magnitudes within  $\sim 0.001$ , the sole exception being `int8_dyn` under category weighting, which was not significant under any convention.

| Aggregation                             | ESM2-650M | ESM2-3B | ESM2-15B |
|-----------------------------------------|-----------|---------|----------|
| Flat mean over 201 assays (this report) | 0.4459    | 0.4398  | 0.4323   |
| Mean of per-protein means               | 0.4403    | 0.4377  | 0.4321   |
| Mean of per-category means              | 0.4297    | 0.4209  | 0.4152   |
| Post-stratified to the full 217         | 0.4406    | 0.4355  | 0.4290   |

and none are stability assays against 32.8% retained. Low-MSA-depth assays score far worse, so the exclusion inflates any unweighted mean — post-stratifying to the full-217 composition costs about 0.004 at 3B. It also *understates* `int4_wo`’s harm at 650M, which moves from  $-0.0064$  to  $-0.0068$  under post-stratification.

Run `python src/analyze_scales.py` to reproduce all four conventions.

## 2.7 Statistical treatment

Ground-truth correlations are compared using a **paired bootstrap** (2000 resamples). Both the candidate configuration and the fp32 reference are scored on the *same* resample at each iteration, so shared noise cancels and the residual is attributable to the configuration. The pairing is not a refinement: for `int4_wo` the paired interval is  $8.4\times$  narrower than the unpaired one, because per-assay  $\rho$  spans 0.0–0.9 while the configuration effect is of order 0.006. Unpaired, every configuration reads as noise.

The *resampling unit* differs by scale of claim. For a single assay the unit is the variant, which establishes that a shift is not an artifact of the variant set. For the full benchmark the unit is the assay, since the claim becomes one about assays in general. Because the 201 assays cover only 173 distinct proteins, benchmark-level intervals use a **cluster bootstrap over proteins**: whole proteins are drawn and all their assays retained, so correlated replicates cannot count as independent evidence. Clustered and unclustered intervals agreed to the fourth decimal, which is reported as a measured result rather than assumed.

Intervals are percentile bootstrap. They were checked against a normal approximation ( $\bar{d} \pm 1.96 \text{ SE}$ ) and agree to four decimals on every configuration, confirming the underlying distribution is near-symmetric and the percentile method is not distorting the interval. The procedure was validated on synthetic data with a known answer: a negligible perturbation was correctly classified as noise, a genuine improvement as real.

## 2.8 Measurement repeatability

Every difference reported here is at  $10^{-4}$  resolution or finer, and the tail statistic is a *maximum* over 201 assays — a quantity any noise inflates, since the maximum of many noisy draws is large even when every true effect is zero. Neither is interpretable without knowing what the pipeline does when nothing changes, so we measured it rather than assuming determinism.

The complete benchmark was re-run for all six configurations at 650M and for `bf16` and `int8_dyn` at 3B, as separate jobs on different physical GPUs. Across 8 model/configuration replicates and 19,311,304 per-variant score comparisons, **every score is identical** at the  $10^{-6}$  precision to which

scores are stored. Per-assay Spearman is unchanged to five decimals, benchmark means to six, and the largest per-assay  $|\Delta\rho|$  between two runs of the same configuration is 0.00000.

So the null for the tail statistic is exactly zero — there is no run-to-run variability for a maximum to select from, and a worst-assay  $\Delta\rho$  of  $-0.369$  is entirely attributable to the configuration. Differences of 0.0002 are properties of the configuration rather than of the run. And the headline collapse reproduces exactly: UBR5\_HUMAN\_Tsuboyama\_2023\_1I2T gives  $\rho = 0.2226$  under `int8_dyn` at 3B in both runs.

Caveats. Determinism removes run-to-run variance and nothing else; the assay-sampling uncertainty the bootstrap quantifies is untouched, and a reproducible 0.0002 is still negligible. Both runs used H200 GPUs and one software stack, so determinism across GPU architectures or torch/torchao versions is not established. Scores are stored rounded to six decimals, so “identical” means identical to  $10^{-6}$  — three orders below the smallest effect discussed.

Reproduce with `python src/analyze_repeat.py`.

## 2.9 Hardware and software environment

| Component       | Detail                                        |
|-----------------|-----------------------------------------------|
| GPU (primary)   | NVIDIA H200, 143 GB, sm_90 (FP8 tensor cores) |
| GPU (secondary) | NVIDIA A100, 80 GB, sm_80 (no FP8)            |
| Driver          | 550.54.14                                     |
| PyTorch         | 2.6.0+cu124                                   |
| Attention       | SDPA                                          |
| Compile mode    | <code>max-autotune-no-cudagraphs</code>       |

The cluster runs CentOS 7.6 with glibc 2.17 on every node, including the H200 nodes, which forced several choices. PyTorch moved to `manylinux_2.28` at version 2.7, making **2.6.0 the newest installable release**. `bitsandbytes` GPU wheels require a newer glibc, so only the CPU-only 0.42.0 installs and its NF4 path is unavailable — 4-bit quantization therefore goes through `torchao`. The system GCC (4.8.5) cannot build Triton’s helper module, so `torch.compile` fails until CC/CXX are pointed at GCC 12.1.0. Finally, `$HOME` and `/project` are at quota, and the Triton, Inductor, HuggingFace, and pip caches all default there; all were relocated to `/scratch`. `torchao` nonetheless works because its INT8/FP8 paths dispatch to `torch._int_mm` and `torch._scaled_mm`, which live inside PyTorch and support sm\_90.

## 3 Results

Sections 3.1–3.6 establish the speed, memory and fidelity behaviour of each configuration on the two workloads, taking ground truth from three hand-picked assays — enough to show that an effect exists on a particular assay, not enough to generalise from it. Section 3.7 repeats the accuracy measurement on the complete 201-assay benchmark at three model scales and overturns two conclusions the three-assay pilot supported; each retraction is stated where the original claim was made.

### 3.1 Compile mode determines whether W8A8 is viable at all

The single most consequential configuration choice is not the quantization scheme but the `torch.compile` mode. Measured on a single ESM2-3B-shaped feed-forward block,  $2560 \rightarrow 10240 \rightarrow 2560$ , at batch  $32 \times 512$  on an A100:

Table 4: Only `max-autotune` allows Inductor to autotune the INT8 Triton matmuls, which is where the entire W8A8 benefit resides.

| Variant                                      | Time (ms)   |
|----------------------------------------------|-------------|
| bf16 eager                                   | 7.91        |
| bf16 compiled                                | 7.91        |
| int8_dyn eager                               | 33.35       |
| int8_dyn compiled, <code>dynamic=True</code> | 11.73       |
| int8_dyn compiled, default mode              | 11.48       |
| int8_dyn mode=" <code>max-autotune</code> "  | <b>5.43</b> |

Measured in eager mode, W8A8 appears  $4.2\times$  *slower* than bf16 and would be discarded. Measured under `max-autotune` it is  $1.46\times$  faster. `dynamic=True` was avoided because it emits shape-generic kernels that surrender most of the gain; static shapes are kept manageable by padding to a fixed multiple of 128. `torch.dynamo.explain` confirms **zero graph breaks** for both bf16 and `int8_dyn`, establishing this as a kernel-selection effect rather than a Dynamo fallback.

### 3.2 Bulk embedding extraction

Table 5: ESM2-3B, H200, compiled with `max-autotune-no-cudagraphs`, SDPA attention, 64 sequences of length 512–1024 under a 16,384-token budget. Reproduced across five independent jobs: bf16 mean 56,789 residues/s, range 55,941–57,422 ( $\pm 1.3\%$ ); fp32 mean 6,752, range 6,687–6,814. The `int8_dyn_async` row is from the fifth job at identical settings, whose bf16 (57,377) and `int8_dyn` (60,023) fall inside those ranges.

| Config         | Weights (GB) | Peak (GB)   | Residues/s    | vs bf16                        | Emb. cos | Logit KL          | DMS $\rho_{\text{fp32}}$ |
|----------------|--------------|-------------|---------------|--------------------------------|----------|-------------------|--------------------------|
| fp32 (ref)     | 10.72        | 25.37       | 6,750         | $0.12\times$                   | —        | —                 | —                        |
| bf16           | 5.29         | 12.63       | 57,057        | $1.00\times$                   | 0.999949 | $1.17\text{e-}04$ | 0.9998                   |
| int8_wo        | 2.66         | 8.48        | 54,816        | $0.96\times$                   | 0.999959 | $1.05\text{e-}04$ | 0.9998                   |
| int8_dyn       | 2.65         | <b>7.07</b> | 59,996        | $1.05\times$                   | 0.989592 | $1.40\text{e-}02$ | 0.9928                   |
| int8_dyn_async | 2.65         | 7.26        | 51,383        | $0.90\times$                   | 0.999469 | $1.41\text{e-}03$ | 0.9972                   |
| fp8_dyn        | 2.65         | 7.24        | <b>65,376</b> | <b><math>1.15\times</math></b> | 0.999920 | $4.60\text{e-}04$ | 0.9988                   |
| int4_wo        | <b>1.61</b>  | 7.69        | 3,813         | $0.07\times$                   | 0.998904 | $1.89\text{e-}03$ | 0.9975                   |

Three observations. First, the **fp32  $\rightarrow$  bf16 step is worth**  $8.4\times$  — larger than everything quantization adds on top of it, and the single biggest lever available. Second, **fp8\_dyn** dominates **int8\_dyn**: it is faster ( $1.15\times$  vs  $1.05\times$ ) *and* dramatically more accurate (embedding cosine 0.999920 vs 0.989592; logit KL  $4.6 \times 10^{-4}$  vs  $1.4 \times 10^{-2}$ ), surrendering only 0.17 GB of peak memory. FP8’s E4M3 format has far greater dynamic range than INT8 at the same bit width, which matters because transformer activations contain outliers. Third, **int4\_wo** is a trap for inference:  $0.07\times$  throughput and *worse* peak memory (7.69 GB) than **fp8\_dyn** (7.24 GB) despite 40% smaller weights, because the dequantization path allocates transient buffers. Its value lies elsewhere, as a QLoRA base.

Separately, length-bucketed batching reduced padding waste from 25.0% (naive fixed-size batching) to **0.0%**. This is free and costs no accuracy — a larger effect than several of the quantization differences in the table.

Table 6: DMS scoring throughput, ESM2-3B, uncompiled. bf16 wins at every problem size; quantization is pure overhead in this regime.

| Config   | Variants/s     |                  |                   |
|----------|----------------|------------------|-------------------|
|          | IF1 ( $L=72$ ) | BLAT ( $L=286$ ) | HSP82 ( $L=709$ ) |
| bf16     | <b>6,979</b>   | <b>3,573</b>     | <b>1,357</b>      |
| int8_wo  | 4,830          | 2,511            | 1,066             |
| fp8_dyn  | 1,813          | 1,796            | 1,103             |
| fp32     | 1,460          | 447              | 187               |
| int8_dyn | 410            | 382              | 299               |
| int4_wo  | 1,045          | 282              | 112               |

### 3.3 DMS scoring: the opposite conclusion

bf16 is fastest at every assay size, and int8\_dyn — the second-fastest configuration for bulk extraction — is up to  $17\times$  slower here. The masked batch ( $n_{\text{positions}} \times L$ ) is too small to contain a matrix multiply large enough for low precision to pay for its quantize/dequantize overhead.

Notably, fp8\_dyn closes on bf16 as the assay grows:  $0.26\times$ ,  $0.50\times$ ,  $0.81\times$  of bf16 throughput at  $L = 72, 286, 709$  respectively. Its overhead is fixed per call while real work scales, so at  $L = 709$  it costs 20% throughput for 43% less memory — a defensible trade that is indefensible at  $L = 72$ .

### 3.4 Memory during DMS scoring

Table 7: Peak device memory during DMS scoring. Weights account for approximately 90% of peak even at  $L = 709$  (bf16: 5.29 GB weights against 0.59 GB activations).

| Config   | Weights (GB) | Peak $L=72$ | Peak $L=286$ | Peak $L=709$ |
|----------|--------------|-------------|--------------|--------------|
| fp32     | 10.72        | 10.87       | 11.19        | 11.85        |
| bf16     | 5.29         | 5.38        | 5.55         | 5.88         |
| int8_wo  | 2.66         | 2.77        | 2.92         | 3.25         |
| int8_dyn | 2.65         | 2.87        | 2.91         | 3.36         |
| fp8_dyn  | 2.65         | 2.76        | 2.96         | 3.35         |
| int4_wo  | <b>1.61</b>  | <b>1.70</b> | <b>1.87</b>  | <b>2.20</b>  |

This is the reverse of the bulk-extraction situation. Masked-marginals scoring keeps the batch narrow, so the  $O(L^2)$  attention term never dominates and weights remain roughly 90% of peak memory even at  $L = 709$ . Quantization therefore addresses almost all of DMS peak memory, and int4\_wo gives the smallest footprint at every length —  $2.7\times$  below bf16.

### 3.5 Ground-truth accuracy on ProteinGym

Two findings emerge, and the second qualifies the first.

**Quantization does not systematically degrade variant-effect accuracy.** Of the 15 assay/configuration pairs, 11 shifts are upward. Most are statistically indistinguishable from zero. Strikingly, int4\_wo — the worst configuration on every fidelity metric — produces the *best* agreement with experiment on BLAT ( $+0.0495$ , CI  $[+0.0433, +0.0561]$ ). This survives the bootstrap and is not a resampling artifact.

Table 8: Spearman correlation against experimental measurements, ESM2-3B.  $\Delta$  is the change from fp32, with a 2000-sample paired bootstrap over variants. “Real” indicates the 95% confidence interval excludes zero.

| Assay                  | Config   | $\rho_{\text{expt}}$ | $\Delta$ vs fp32 | 95% CI of $\Delta$ | Verdict     |
|------------------------|----------|----------------------|------------------|--------------------|-------------|
| IF1<br>( $n=1367$ )    | fp32     | 0.5561               | (ref)            |                    |             |
|                        | bf16     | 0.5543               | −0.0018          | [−0.0027, −0.0009] | <b>real</b> |
|                        | int8_wo  | 0.5570               | +0.0009          | [−0.0003, +0.0021] | noise       |
|                        | int8_dyn | 0.5504               | −0.0057          | [−0.0124, +0.0013] | noise       |
|                        | fp8_dyn  | 0.5584               | +0.0023          | [−0.0022, +0.0067] | noise       |
|                        | int4_wo  | 0.5394               | <b>−0.0167</b>   | [−0.0263, −0.0074] | <b>real</b> |
| BLAT<br>( $n=4996$ )   | fp32     | 0.5893               | (ref)            |                    |             |
|                        | bf16     | 0.5896               | +0.0003          | [−0.0003, +0.0009] | noise       |
|                        | int8_wo  | 0.5913               | +0.0019          | [+0.0011, +0.0029] | <b>real</b> |
|                        | int8_dyn | 0.6160               | <b>+0.0266</b>   | [+0.0219, +0.0315] | <b>real</b> |
|                        | fp8_dyn  | 0.6088               | <b>+0.0194</b>   | [+0.0161, +0.0229] | <b>real</b> |
|                        | int4_wo  | 0.6388               | <b>+0.0495</b>   | [+0.0433, +0.0561] | <b>real</b> |
| HSP82<br>( $n=13194$ ) | fp32     | 0.2931               | (ref)            |                    |             |
|                        | bf16     | 0.2933               | +0.0002          | [−0.0002, +0.0005] | noise       |
|                        | int8_wo  | 0.2933               | +0.0002          | [−0.0002, +0.0007] | noise       |
|                        | int8_dyn | 0.2937               | +0.0006          | [−0.0020, +0.0028] | noise       |
|                        | fp8_dyn  | 0.2944               | +0.0012          | [−0.0003, +0.0027] | noise       |
|                        | int4_wo  | 0.2904               | −0.0027          | [−0.0057, +0.0002] | noise       |

**Fidelity-vs-fp32 predicts the magnitude of the shift, not its sign.** Across all 15 pairs, the correlation between fidelity loss ( $1 - \rho_{\text{fp32}}$ ) and the absolute ground-truth shift is  $r = 0.87$  (Pearson),  $\rho = 0.76$  (Spearman). But **int4\_wo** moves +0.0495 on BLAT and −0.0167 on IF1. The direction is a property of the assay, not of the configuration, and is not predictable from any drift-from-fp32 measurement.

The practical consequence is that **bf16** and **int8\_wo** shift ground-truth agreement by at most 0.002 on any assay tested — below anything an experiment could resolve — and are therefore safe. **int4\_wo** is a  $\pm 0.05$  gamble on an unknown sign.

This supersedes an earlier conclusion drawn from a synthetic 40-residue mutational scan, which ranked **int8\_dyn** ( $\rho_{\text{fp32}} = 0.9928$ ) as risky and **fp8\_dyn** (0.9988) as clearly safer. Both numbers were correct; neither predicted experimental accuracy. On BLAT the “risky” configuration gained +0.027.

We note that the assay itself matters far more than any quantization decision:  $\rho_{\text{expt}}$  ranges from 0.29 (HSP82) to 0.59 (BLAT) at 3B. Choosing which assay to trust is a  $2\times$  effect; choosing a quantization configuration is a 1% effect.

### 3.6 Model-scale consistency

The same three assays were run on ESM2-650M. The qualitative picture holds: **bf16** fastest, **int8\_dyn** slowest, ground-truth shifts small. The largest 650M shift was −0.0063 (**int8\_dyn** on IF1) against  $\rho_{\text{fp32}}$  of 0.9954. The larger and more clearly significant shifts at 3B appear to indicate that sensitivity to quantization grows with model size.

*That reading does not survive the third scale.* At 15B no configuration shows a significant effect at all (Table 9), so sensitivity does not increase monotonically with size; it is simply not a function of size in any usable way.



The pilot appeared to say something stronger as well. The fp32 650M model correlated with experiment *better* than fp32 3B on two of the three assays, and on BLAT by 0.1422 (0.7315 against 0.5893) — roughly *three times* the largest effect any quantization configuration produced in this study. The inference drawn was that model selection dominates precision selection by a wide margin.

*That inference does not survive Section 3.7.* Over 201 assays the two models are statistically indistinguishable ( $p = 0.35$ ), and a third scale makes the picture worse for the larger checkpoint rather than better: fp32 mean  $\rho$  falls monotonically  $0.4459 \rightarrow 0.4398 \rightarrow 0.4323$  from 650M to 3B to 15B, with no pairwise difference significant. The BLAT gap is real for BLAT and does not generalise — the same error, made the same way, as the `int4_wo` result three subsections above.

### 3.7 The full benchmark at three model scales

Every accuracy conclusion above rests on three assays and one model. Three assays establish that an effect exists on a particular assay, not anything about assays in general, because the unit that varies is held fixed. One model scale establishes that an effect exists for that checkpoint, not that it transfers. The whole ProteinGym substitution benchmark was therefore run at three scales: all 217 assays validated, the 201 within ESM-2’s 1022-residue context scored, 2,413,913 variants over 40,775 masked positions, six configurations, at 650M, 3B and 15B. 3618 assay/configuration runs, 16.2 GPU-hours.

The resampling unit becomes the **assay** rather than the variant, and because the 201 assays cover only 173 distinct proteins the reported interval is a **cluster bootstrap over proteins** — whole proteins are drawn and all their assays retained, so correlated replicates cannot count as independent evidence. Clustered and unclustered intervals agree to the fourth decimal, which is reported as a measured result rather than assumed.

Table 9: Change in mean Spearman against experiment relative to fp32, over 201 assays, at each model scale. Bold marks intervals excluding zero. `int4_wo` is significant in *opposite directions* at 650M and 3B and null at 15B: the effect has no consistent sign and no monotone trend in scale. fp32 mean  $\rho$  is 0.4459, 0.4398, 0.4323 respectively.

| Config                | ESM2-650M      |                    | ESM2-3B        |                    | ESM2-15B |                    |
|-----------------------|----------------|--------------------|----------------|--------------------|----------|--------------------|
|                       | $\Delta$       | 95% CI             | $\Delta$       | 95% CI             | $\Delta$ | 95% CI             |
| <code>bf16</code>     | +0.0000        | [−0.0003, +0.0004] | +0.0002        | [−0.0001, +0.0004] | +0.0000  | [−0.0009, +0.0011] |
| <code>int8_wo</code>  | −0.0002        | [−0.0006, +0.0002] | +0.0001        | [−0.0002, +0.0005] | +0.0002  | [−0.0008, +0.0014] |
| <code>fp8_dyn</code>  | −0.0008        | [−0.0020, +0.0005] | +0.0005        | [−0.0007, +0.0017] | +0.0002  | [−0.0010, +0.0015] |
| <code>int8_dyn</code> | <b>−0.0020</b> | [−0.0032, −0.0009] | −0.0021        | [−0.0072, +0.0017] | −0.0000  | [−0.0021, +0.0019] |
| <code>int4_wo</code>  | <b>−0.0064</b> | [−0.0105, −0.0018] | <b>+0.0037</b> | [+0.0007, +0.0068] | +0.0009  | [−0.0014, +0.0034] |

#### The benchmark mean is the wrong safety criterion

Every  $\Delta$  in Table 9 is below 0.007. The per-assay tail is not.

`int8_dyn` at 3B is indistinguishable from fp32 on the benchmark mean ( $\Delta = -0.0021$ ,  $p = 0.34$ ) while destroying `UBR5_HUMAN.Tsuboyama_2023.1I2T`, where correlation with experiment falls from 0.591 to 0.223 at  $\rho_{\text{fp32}} = 0.393$ . A configuration can pass the benchmark average and be unusable on the one protein a project cares about. Knowing which configuration was dangerous at one scale also does not tell you which will be dangerous at another.

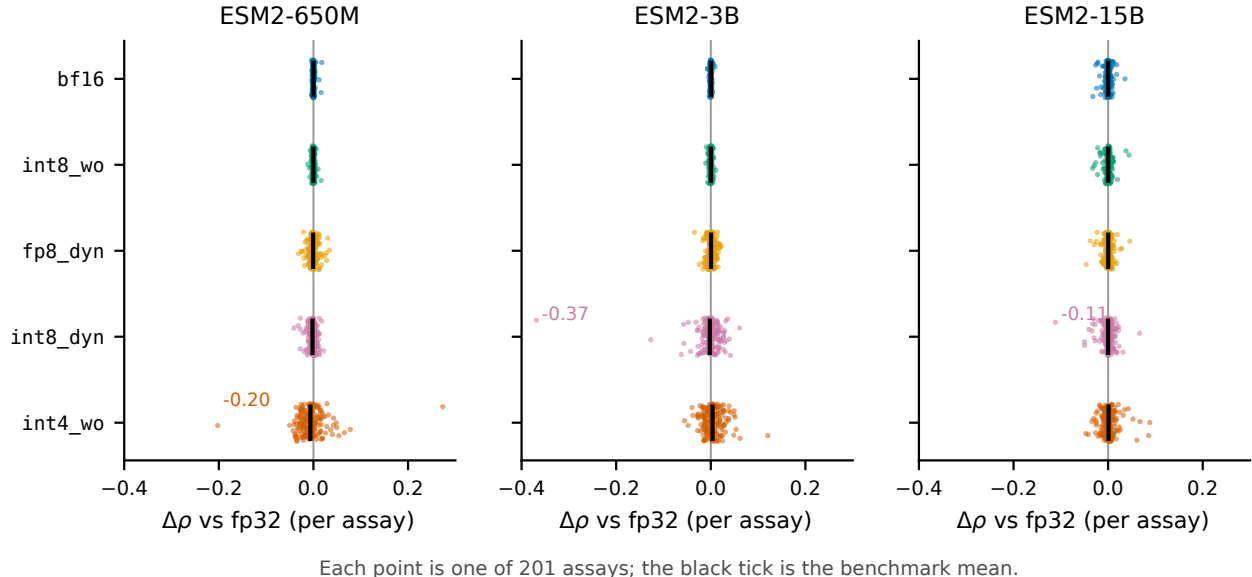


Figure 1: Per-assay change in ground-truth correlation, all 201 assays, all three scales. Black ticks are the benchmark mean, within 0.007 of zero everywhere. `bf16` and `int8_wo` are tight at every assay and scale; `int8_dyn` and `int4_wo` have equally unremarkable means and tails reaching  $-0.37$  and  $-0.20$ .

Table 10: Worst single-assay change in ground-truth correlation, with the count of assays moving more than 0.05 in parentheses, out of 201. The catastrophic tail does not belong to a fixed configuration: `int4_wo` at 650M, `int8_dyn` at 3B.

| Config                | ESM2-650M                       | ESM2-3B                         | ESM2-15B      |
|-----------------------|---------------------------------|---------------------------------|---------------|
| <code>bf16</code>     | $-0.0040$ (0)                   | $-0.0072$ (0)                   | $-0.0323$ (0) |
| <code>int8_wo</code>  | $-0.0120$ (0)                   | $-0.0106$ (0)                   | $-0.0329$ (0) |
| <code>fp8_dyn</code>  | $-0.0316$ (0)                   | $-0.0344$ (0)                   | $-0.0465$ (0) |
| <code>int8_dyn</code> | $-0.0411$ (0)                   | <b><math>-0.3688</math></b> (6) | $-0.1112$ (3) |
| <code>int4_wo</code>  | <b><math>-0.2024</math></b> (6) | $-0.0556$ (5)                   | $-0.0472$ (5) |

### At 15B the safe configurations degrade and the aggressive one improves

Scale does not move all configurations in the same direction, and the crossover needs a third point to see (Figure 2c, Table 11). `bf16` and `int8_wo` do not drift measurably from `fp32` on *any* of the 402 assay/model pairs at 650M and 3B, then drift on 13 and 15 assays at 15B, with worst-case rank correlation falling to 0.857 and 0.855. `int4_wo` moves the other way: 182 assays below  $\rho_{\text{fp32}} = 0.99$  at 650M, 154 at 3B, 82 at 15B.

A natural reading is that a larger model carries more redundancy, so removing weight precision costs proportionally less, while its activation dynamic range grows until `bf16`’s exponent becomes the binding constraint. We cannot assert that mechanism from this data — it would need a per-layer error analysis and an `fp16`-versus-`bf16` comparison, neither of which we ran. What the data does support is that **the identity of the risky configuration is not stable across scale**, so clearing a configuration at one scale does not clear it at another. The 15B `bf16` degradation is also not a long-sequence artifact: Spearman correlation between assay length and  $\rho_{\text{fp32}}$  is  $-0.14$ , and the five worst assays span  $L = 222$  to 861.

Table 11: Fidelity against fp32 by scale: assays with  $\rho_{\text{fp32}} < 0.99$  (of 201), and the worst single-assay value. The configurations that are safe at small scale acquire a tail at 15B; the most aggressive one loses its tail. Medians stay above 0.99 everywhere, so none of this shows in a summary statistic.

| Config   | assays with $\rho_{\text{fp32}} < 0.99$ |     |     | worst $\rho_{\text{fp32}}$ |        |        |
|----------|-----------------------------------------|-----|-----|----------------------------|--------|--------|
|          | 650M                                    | 3B  | 15B | 650M                       | 3B     | 15B    |
| bf16     | 0                                       | 0   | 13  | 0.9974                     | 0.9905 | 0.8569 |
| int8_wo  | 0                                       | 0   | 15  | 0.9963                     | 0.9968 | 0.8548 |
| fp8_dyn  | 29                                      | 18  | 19  | 0.9556                     | 0.9770 | 0.8389 |
| int8_dyn | 25                                      | 108 | 31  | 0.8894                     | 0.3930 | 0.7645 |
| int4_wo  | 182                                     | 154 | 82  | 0.6707                     | 0.8791 | 0.8505 |

Table 12: The single worst-affected assay, by configuration and scale. No assay appears twice: identifying the vulnerable target at one scale does not identify it at another.

| Scale | Config   | Worst assay                                 | $\Delta\rho$ |
|-------|----------|---------------------------------------------|--------------|
| 650M  | bf16     | AICDA_HUMAN_Gajula_2014_3cycles ( $L=198$ ) | -0.0040      |
| 650M  | int8_dyn | POLG_PESV_Tsuboyama_2023_2MXD ( $L=53$ )    | -0.0411      |
| 650M  | int4_wo  | B2L11_HUMAN_Dutta_2010 ( $L=198$ )          | -0.2024      |
| 3B    | bf16     | ODP2_GEOSE_Tsuboyama_2023_1W4G ( $L=44$ )   | -0.0072      |
| 3B    | int8_dyn | UBR5_HUMAN_Tsuboyama_2023_1I2T ( $L=58$ )   | -0.3688      |
| 3B    | int4_wo  | R1AB_SARS2_Flynn_2022 ( $L=306$ )           | -0.0556      |
| 15B   | bf16     | I6TAH8_I68A0_Doud_2015 ( $L=498$ )          | -0.0323      |
| 15B   | int8_dyn | GFP_AEQVI_Sarkisyan_2016 ( $L=238$ )        | -0.1112      |
| 15B   | int4_wo  | MBD11_ARATH_Tsuboyama_2023_6ACV ( $L=66$ )  | -0.0472      |

## Resource envelope at 15B

### The configuration ranking is scale-dependent

Section 5.1 reports that the two workloads prefer opposite configurations. That holds at 650M and 3B and stops holding at 15B.

At 15B, fp8\_dyn scores the benchmark in 0.381 GPU-hours against bf16’s 0.385, at peak memory 14.57 against 28.57 GB, with no measurable accuracy cost. It is simply the better choice, which it is not at either smaller scale. The workload conflict is a small-model phenomenon and a recommendation derived at 650M or 3B does not transfer upward.

### What the larger sample changed

**The three-assay verdict was backwards, and the correction does not resolve into a trend.** Section 3.5 concluded that quantization does not systematically degrade variant-effect accuracy, on the evidence that 11 of 15 shifts were upward and int4\_wo gained +0.0495 on BLAT. At benchmark scale int4\_wo is significantly harmful at 650M, significantly beneficial at 3B, and null at 15B. Two scales would have left this looking like a trend in model size; the third removes that reading.

**Fidelity predicts magnitude, not direction.** Across 3015 assay/configuration pairs,  $r(1 - \rho_{\text{fp32}}, |\Delta|)$  is 0.74, 0.81 and 0.56 at 650M, 3B, 15B — consistent with the 0.87 first estimated from 15 pairs. The *signed* correlation is +0.10, -0.58, -0.40: it does not hold its sign across scales and is useless for prediction.

Table 13: ESM2-15B footprint. Peak memory during masked-marginals scoring is almost entirely weights: the spread across 201 assays spanning  $L = 37$  to 934 is under 3 GB for every configuration. Quantization therefore addresses essentially all of the DMS memory cost at this scale, and fp32 alone rules out an 80 GB A100. Model load takes about six minutes per configuration from the shared filesystem, which is why the driver loads once and scores all 201 assays.

| Config          | Weights (GB) | Peak memory over 201 assays (GB) |        |       |
|-----------------|--------------|----------------------------------|--------|-------|
|                 |              | min                              | median | max   |
| <b>fp32</b>     | 56.36        | 56.51                            | 57.09  | 59.30 |
| <b>bf16</b>     | 28.18        | 28.27                            | 28.57  | 29.69 |
| <b>int8_wo</b>  | 14.14        | 14.39                            | 14.53  | 15.64 |
| <b>fp8_dyn</b>  | 14.12        | 14.22                            | 14.57  | 15.91 |
| <b>int8_dyn</b> | 14.12        | 14.51                            | 14.90  | 16.06 |
| <b>int4_wo</b>  | 7.53         | 7.62                             | 7.91   | 9.03  |

Table 14: DMS scoring speed relative to **bf16** over the full benchmark, by scale. Weight-only **int8\_wo** holds a constant ratio, as expected for a scheme adding fixed dequantization work. **fp8\_dyn** closes from  $0.30\times$  to  $0.97\times$ : its overhead is per-call while useful work grows with hidden size, so by 15B the GEMMs are compute-bound even in the narrow-batch DMS regime.

| Config          | ESM2-650M |               | ESM2-3B |               | ESM2-15B |                                |
|-----------------|-----------|---------------|---------|---------------|----------|--------------------------------|
|                 | pos/s     | vs bf16       | pos/s   | vs bf16       | pos/s    | vs bf16                        |
| <b>bf16</b>     | 347.7     | 1.00 $\times$ | 200.6   | 1.00 $\times$ | 56.7     | 1.00 $\times$                  |
| <b>int8_wo</b>  | 249.0     | 0.72 $\times$ | 139.3   | 0.69 $\times$ | 41.0     | 0.72 $\times$                  |
| <b>fp8_dyn</b>  | 103.3     | 0.30 $\times$ | 89.2    | 0.44 $\times$ | 54.9     | <b>0.97<math>\times</math></b> |
| <b>int8_dyn</b> | 23.8      | 0.07 $\times$ | 20.1    | 0.10 $\times$ | 11.3     | 0.20 $\times$                  |
| <b>int4_wo</b>  | 55.4      | 0.16 $\times$ | 17.3    | 0.09 $\times$ | 3.9      | 0.07 $\times$                  |

**Stratified effects do not replicate either.** At 650M, **int4\_wo**’s cost is monotone in MSA depth ( $-0.0174$ ,  $-0.0050$ ,  $-0.0042$  for low, medium, high), inviting the reading that quantization harms most where the model has least evolutionary signal. That ordering holds at 3B ( $-0.0011$ ,  $+0.0007$ ,  $+0.0097$ ) and breaks at 15B ( $+0.0011$ ,  $-0.0009$ ,  $+0.0033$ ), where every effect is inside the noise band anyway. Two scales agreeing and a third not is exactly the pattern the **int4\_wo** result should teach us to distrust.

### The W8A8 collapse is a scaling defect, not a limit of the scheme

“**int8\_dyn** destroys an assay” and “W8A8 cannot work for ESM-2” carry very different recommendations, so we tested which one the data supports.

Coarse granularity is not available as an explanation: **int8\_dyn** already uses per-token activation and per-channel weight scales. We profiled per-channel activation magnitudes at every encoder Linear input on the collapsing assay and four length-matched controls, and **found no distinguishing signature** — outlier ratio median 16.8 against controls 15.5–17.2, and a largest raw activation of 14.8 against controls 14.9–17.5, i.e. smaller. That negative result is about the diagnostic as much as the mechanism: aggregating over 180 Linears washes out a defect localised to a few.

The intervention test is decisive where the diagnostic was not.

Asymmetric scaling is the better remedy: no calibration, no prototype dependency, and  $1.95\times$  faster than symmetric **int8\_dyn** in our measurements (20.6 against 40.0 minutes) at 0.6 GB more

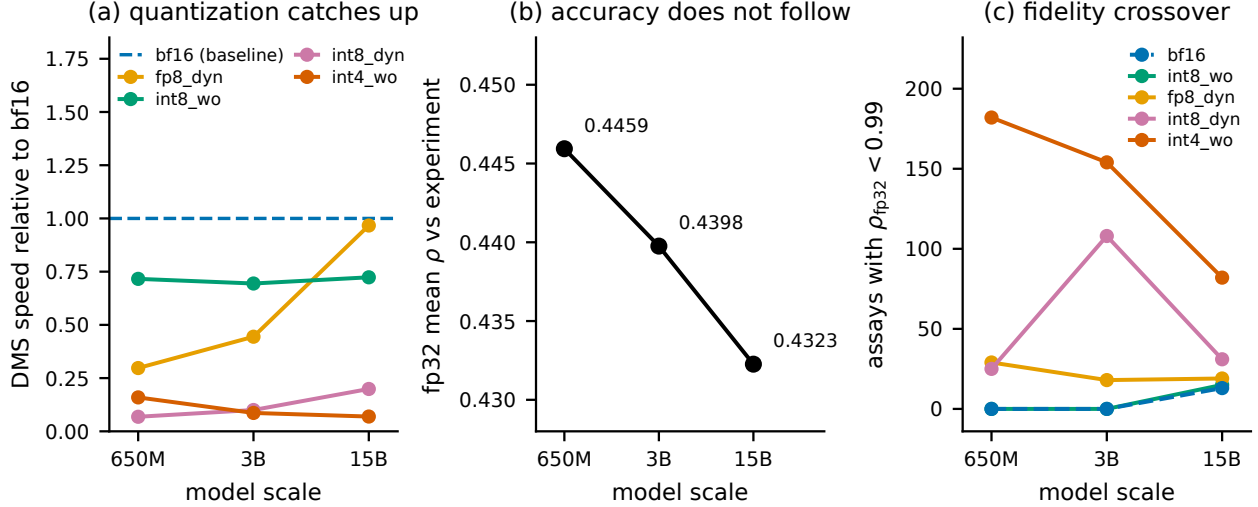


Figure 2: Three effects visible only across scale. (a) DMS speed relative to bf16: fp8\_dyn converges on bf16 while the weight-only and INT4 paths do not. (b) fp32 mean Spearman falls monotonically across a 23-fold parameter increase; no pairwise difference is significant, but there is no evidence that scale buys variant-effect accuracy. (c) Number of assays where a configuration drifts from fp32 ( $\rho_{fp32} < 0.99$ ): bf16 and int8\_wo acquire a tail only at 15B while int4\_wo sheds one.

Table 15: Repairing W8A8 at 3B over the full benchmark. int8\_dyn\_asym is the same config with asymmetric rather than symmetric activation quantization — one keyword, no calibration. int8\_dyn\_sq adds SmoothQuant ( $\alpha = 0.5$ , 16 calibration sequences). Both eliminate all six damaged assays.

| Config        | mean $\Delta$ | worst $\Delta$ | $ \Delta  > 0.05$ | worst $\rho_{fp32}$ | pos/s       | peak GB |
|---------------|---------------|----------------|-------------------|---------------------|-------------|---------|
| int8_dyn      | -0.0021       | <b>-0.3688</b> | 6                 | 0.393               | 17.0        | 3.51    |
| int8_dyn_asym | -0.0001       | -0.0395        | <b>0</b>          | 0.965               | <b>33.1</b> | 4.13    |
| int8_dyn_sq   | +0.0002       | <b>-0.0237</b> | <b>0</b>          | 0.737               | 16.9        | 3.66    |

peak memory. We did not investigate the speed difference. SmoothQuant reaches a slightly better worst case (-0.024 against -0.040) but costs a calibration pass and has *poorer* worst-assay fidelity (0.737 against 0.965).

So “avoid int8\_dyn” was too general. Its default *symmetric* activation scaling is unsafe on ESM-2 and a one-line change repairs it.

### A label-free screen for which assays are at risk

What decides whether a ranking survives a perturbation is the perturbation relative to the spread of the scores being perturbed. Both are computable against fp32 on a user’s own target, with no experimental labels:  $SNR = sd(\sigma_{fp32})/sd(\sigma_{cand} - \sigma_{fp32})$ .

SNR is not merely re-discovering the configuration ranking. Median SNR runs from 47.9 (bf16) to 5.7 (int4\_wo), but holding the configuration fixed it still predicts damage within it — Spearman -0.43, -0.40, -0.26, -0.39, -0.39 for bf16, int8\_wo, fp8\_dyn, int8\_dyn, int4\_wo. Within int4\_wo alone the probability of losing more than 0.05 falls 10.5%  $\rightarrow$  1.6%  $\rightarrow$  0.0% across  $SNR < 4$ , 4–8,  $> 8$ . It carries per-target information.

Reproduce with `python src/analyze_scales.py` (section 5) and `src/diag_outliers.py`.

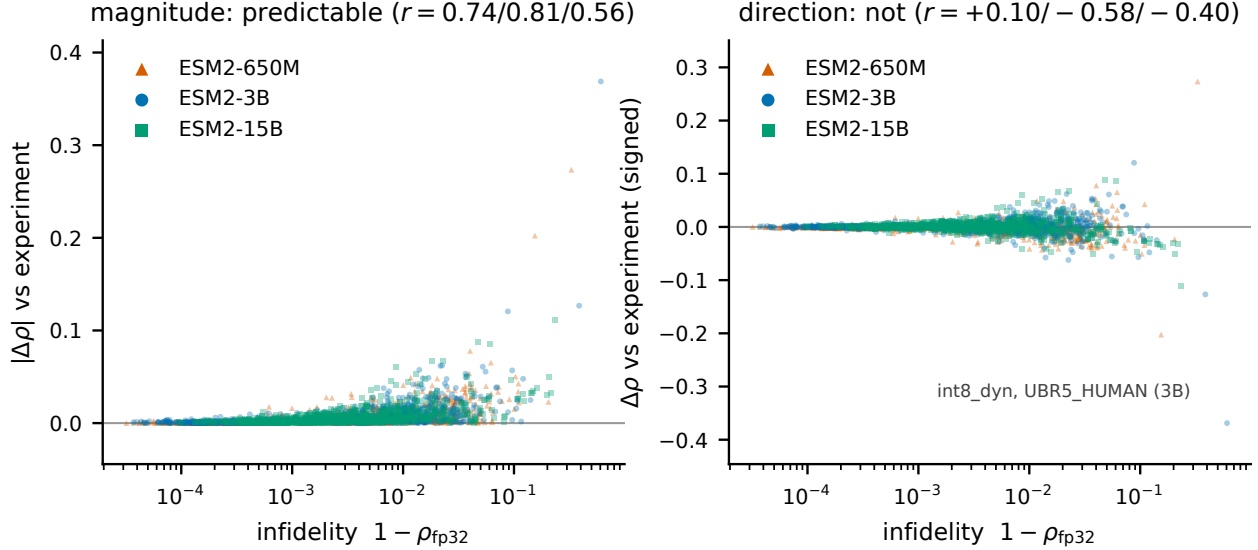


Figure 3: Drift from fp32 against change in ground-truth correlation, 1005 pairs per scale. *Left*: infidelity bounds the magnitude of the change. *Right*: signed, the distribution is a symmetric funnel rather than a trend, and the correlation changes sign between scales.

Table 16: SNR against realised damage over all 3015 assay/configuration/scale combinations. Every collapse beyond 0.10 has SNR < 1.7; nothing with SNR > 8 is damaged beyond 0.05. Usable as an all-clear, not as a precise predictor.

| SNR | $n$  | median $ \Delta $ | $P( \Delta  > 0.05)$ | worst $\Delta$ |
|-----|------|-------------------|----------------------|----------------|
| < 2 | 13   | 0.0478            | 46.2%                | -0.3688        |
| 2-4 | 139  | 0.0165            | 7.2%                 | -0.0569        |
| 4-8 | 630  | 0.0092            | 1.4%                 | -0.0622        |
| > 8 | 2233 | 0.0019            | <b>0.0%</b>          | -0.0409        |

### Quantization competes with using a smaller model, and loses

Every comparison so far is within a scale. That is the wrong frame for a practitioner, who is not obliged to run 3B at all. Placing all three scales on one Pareto surface over (accuracy, peak memory, speed), **only three of eighteen configurations are non-dominated, and all three are 650M** (Figure 6, Table 17).

Quantization remains useful *within* the frontier model: 650M/int8\_wo halves the memory of 650M/bf16 for a 0.0002 accuracy change. But the memory-oriented configurations at 3B and 15B are not trade-offs against a small model; they are strictly worse than one. This holds for variant-effect scoring, where 650M is indistinguishable from 3B and 15B, and does not transfer to tasks where the larger checkpoints are genuinely better.

### Interaction: is the effect actually different between scales?

Calling int4\_wo “significant in opposite directions” rests on two separate tests against fp32 whose verdicts are compared by eye, and those are only nominally significant: 15 comparisons are reported (five configurations, three scales) against a Bonferroni threshold of 0.0033, and the two int4\_wo results give  $p = 0.0102$  and  $p = 0.0104$ .

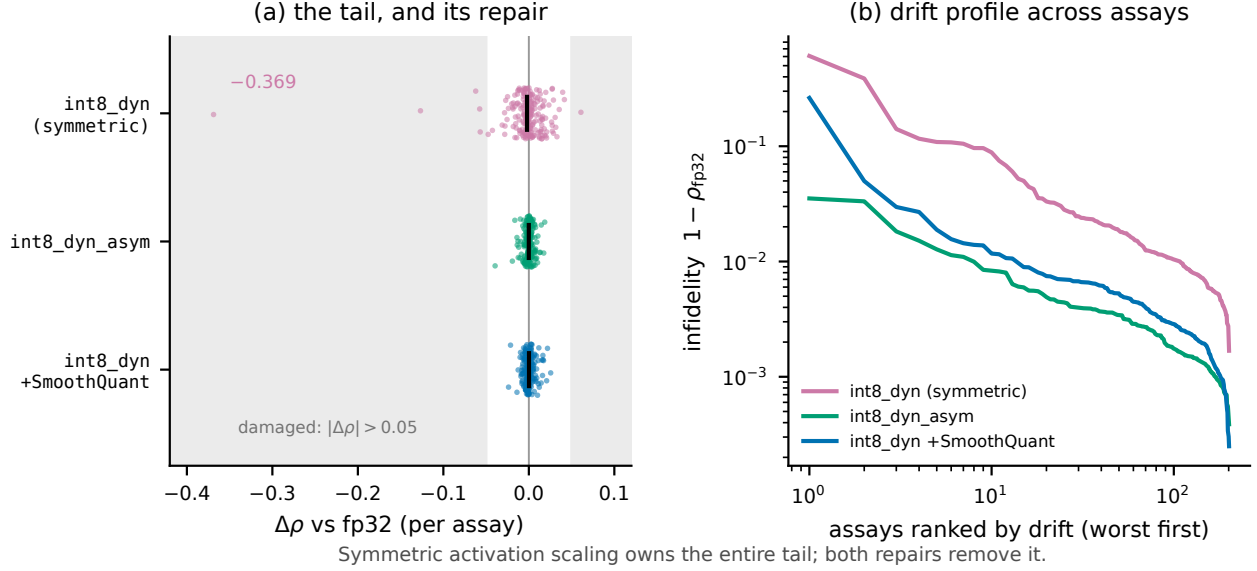


Figure 4: **(a)** Per-assay  $\Delta\rho$  for the three W8A8 variants; shaded bands mark  $|\Delta\rho| > 0.05$ , black ticks the mean. Symmetric scaling owns the entire tail and both repairs remove it while the mean barely moves. **(b)** Drift from fp32, assays ranked worst-first: the separation is not confined to the tail — symmetric is roughly threefold worse at the median.

The claim we want is that the effect *differs* between scales, which is one paired test on the same assays and is the stronger one.

### Conditioning the tail on the reference having signal

A large drop on an assay where fp32 itself scores near zero is rank instability, not damage. Restricting to the 156–157 assays where fp32 reaches  $\rho > 0.3$ : the 3B `int8_dyn` collapse is unchanged at  $-0.369$  (fp32 reaches 0.591 there), while 650M `int4_wo` falls from  $-0.202$  to  $-0.050$  and 15B `int8_dyn` from  $-0.111$  to  $-0.030$ , because those occur where fp32 reaches only 0.270 and 0.290. Across cuts from 0.0 to 0.4 the 3B `int8_dyn` figure is invariant and the 650M `int4_wo` figure is not, so the central example is robust and the  $-0.20$  figure is not.

On assays where the model is usable at all, `int8_dyn` at 3B damages 2 of 157 and `int4_wo` 4–5 of 157 by more than 0.05 — roughly a 1–3% chance a given target is affected.

### Speed and memory over the whole benchmark

The DMS recommendation survives at 650M and 3B and inverts at 15B. Up to 3B, `bf16 eager` is fastest at every problem size —  $6.2\times$  faster than fp32 end-to-end at 3B — with ground-truth agreement never moving more than 0.018 across 402 assay/model pairs, and `int8_wo` is the memory fallback at roughly 70% of bf16 speed. At 15B, `fp8_dyn` matches bf16 on speed at half the memory and becomes the default. `int8_dyn` and `int4_wo` are not trade-offs at any scale: slower than the best option *and* owners of the two worst tails in Table 10.

## 4 Measurement Artifacts

Four measurement defects were found and corrected during this study. Three of them inverted or corrupted the very quantity being measured, and we document them because each is a general

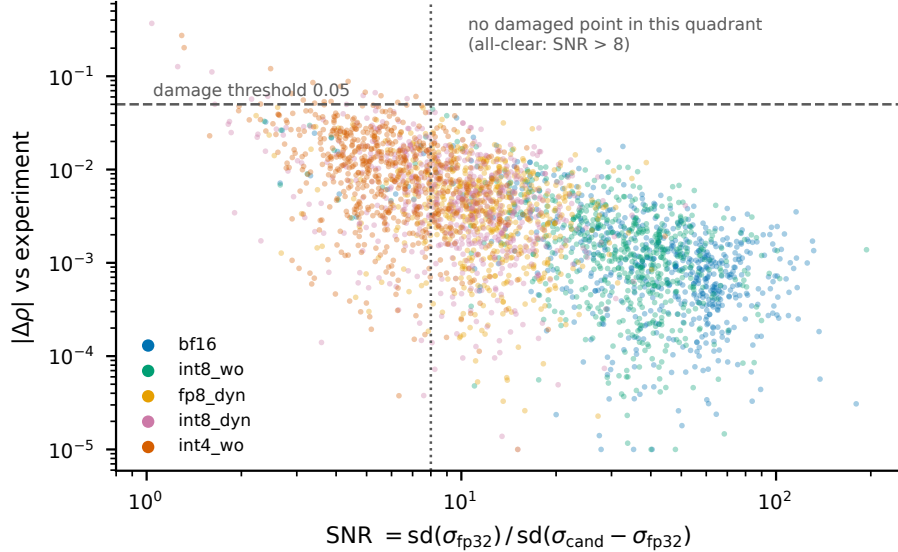


Figure 5: Signal-to-perturbation ratio against realised damage, 3015 combinations. The upper-right quadrant is empty: nothing above  $\text{SNR} = 8$  loses more than 0.05. Colour is the configuration, and their left-to-right ordering recovers their aggressiveness unprompted.

hazard rather than a local mistake.

#### 4.1 Compile cost inside the timed region (three occurrences)

The DMS throughput column initially ranked `fp32` fastest and `fp8_dyn` slowest — an exact inversion of the bulk-throughput column. Two successive fixes failed. Direct instrumentation finally established the cause. On 650M/A100 with a 40-residue wild type over 11 masked positions:

DMS scoring uses a much smaller input shape than the bulk-throughput batches, and under `max-autotune` each new shape costs roughly 15s to autotune. At this batch size that cost recurs rather than amortizing. The timed region was therefore measuring *autotune cost*, and ranking configurations by their number of Triton kernel candidates — `fp32` had the fewest and appeared fastest, `fp8_dyn` had 98 and appeared slowest. The fix was to score DMS against the uncompiled module.

This has a consequence beyond the benchmark: for short wild-type sequences with few mutated positions, `max-autotune` is actively counterproductive in production, not merely in measurement.

The same class of defect appeared earlier in the bulk-throughput path, where warming up on only a prefix of the batches left recompilation for unseen length buckets inside the timed region.

#### 4.2 Peak memory attributed to the wrong model

Peak memory during DMS scoring was reported as follows, with the activation overhead implied by subtracting weights:

Each configuration’s peak included the *previous* configuration’s weights, producing the visible absurdity of `bf16` requiring more peak memory than `fp32`. The cleanup helper deleted only its own local reference to the model; the caller retained a binding. Because `nn.Module` graphs contain reference cycles, dropping the last reference does not free them by reference counting alone — they persist as cyclic garbage until a collection runs. Inserting an explicit `gc.collect()` before each peak-memory reset corrected it; overhead became a consistent 0.04–0.07 GB.



Table 17: Cross-scale Pareto surface. An option is dominated if another is at least as good on accuracy, peak memory *and* speed and strictly better on one. 3B/`int4_wo` — the configuration our recommendations table previously offered for minimum memory — uses more memory than 650M/`bf16` while being less accurate and 20× slower.

| Scale | Config                | mean $\rho$ | peak GB | pos/s | Status                                  |
|-------|-----------------------|-------------|---------|-------|-----------------------------------------|
| 650M  | <code>bf16</code>     | 0.4460      | 1.33    | 347.7 | <b>frontier</b>                         |
| 650M  | <code>int8_wo</code>  | 0.4457      | 0.73    | 249.0 | <b>frontier</b>                         |
| 650M  | <code>int4_wo</code>  | 0.4395      | 0.60    | 55.4  | <b>frontier</b>                         |
| 650M  | <code>fp32</code>     | 0.4459      | 2.70    | 95.0  | dominated by 650M/ <code>bf16</code>    |
| 650M  | <code>fp8_dyn</code>  | 0.4452      | 0.75    | 103.3 | dominated by 650M/ <code>int8_wo</code> |
| 650M  | <code>int8_dyn</code> | 0.4440      | 0.75    | 23.8  | dominated by 650M/ <code>int8_wo</code> |
| 3B    | <code>int4_wo</code>  | 0.4435      | 1.82    | 17.3  | dominated by 650M/ <code>bf16</code>    |
| 3B    | <code>bf16</code>     | 0.4399      | 5.50    | 200.6 | dominated by 650M/ <code>bf16</code>    |
| 3B    | <code>int8_wo</code>  | 0.4399      | 2.87    | 139.3 | dominated by 650M/ <code>bf16</code>    |
| 3B    | <code>fp32</code>     | 0.4398      | 11.10   | 25.6  | dominated by 650M/ <code>fp32</code>    |
| 15B   | <code>int4_wo</code>  | 0.4332      | 7.91    | 3.9   | dominated by 650M/ <code>fp32</code>    |
| 15B   | <code>fp8_dyn</code>  | 0.4325      | 14.57   | 54.9  | dominated by 650M/ <code>fp32</code>    |
| 15B   | <code>bf16</code>     | 0.4323      | 28.57   | 56.7  | dominated by 650M/ <code>fp32</code>    |

Table 18: Interaction contrasts for `int4_wo` at  $R = 20,000$ . The 3B–650M contrast survives Bonferroni correction for all 15 comparisons, where neither individual test does. 15B does not differ from 3B, so the dependence is not monotone.

| Contrast   | Difference | 95% CI             | $p$           |
|------------|------------|--------------------|---------------|
| 3B – 650M  | +0.0101    | [+0.0046, +0.0155] | <b>0.0007</b> |
| 15B – 650M | +0.0073    | [+0.0021, +0.0123] | 0.0067        |
| 15B – 3B   | −0.0028    | [−0.0062, +0.0006] | 0.1017        |

The bulk-throughput memory column was verified unaffected: `bf16`’s measured overhead there was +7.34 GB, which cannot contain `fp32`’s 10.72 GB of weights.

### 4.3 Silent argument leakage in the job script

A batch script used `shift 3 || true`. When fewer than three positional arguments are supplied, `shift 3` fails, `|| true` suppresses the failure, and the arguments remain in "\$@" — where they were appended to the Python invocation as stray arguments. Relying on a default for the third argument was sufficient to trigger it.

### 4.4 Common thread

Three of the four defects shared a signature: a cost that belonged outside the measurement leaked inside it, and the resulting ranking was plausible enough to be believed. In each case the tell was a result that inverted or contradicted a neighbouring measurement. We would emphasise that a benchmark producing a *clean, monotonic, plausible* table is not thereby correct; the memory bug produced exactly such a table for several runs.

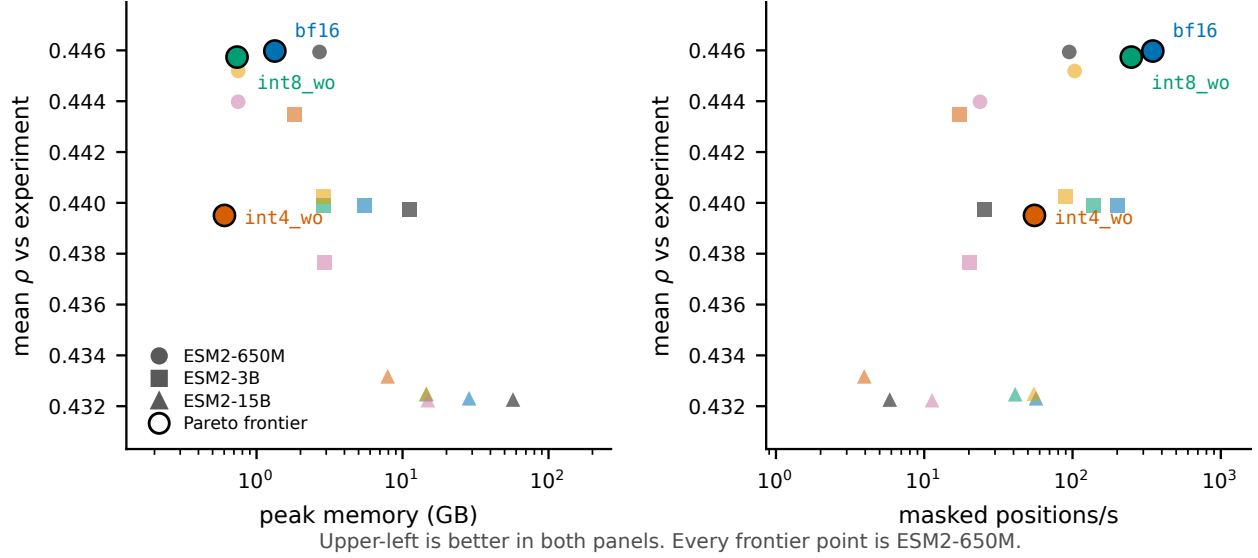


Figure 6: All eighteen scale/configuration combinations on two projections of the Pareto surface; upper-left is better. Circled points are non-dominated and all are ESM2-650M.

Table 19: Full 201-assay benchmark by scale, uncompiled, one H200. Wall-clock is summed scoring time; peak memory is the median across assays. fp32 at 15B needs 57 GB, so the reference configuration alone rules out an 80 GB A100 once activations are included.

| Config   | ESM2-650M   |             | ESM2-3B     |             | ESM2-15B    |             |
|----------|-------------|-------------|-------------|-------------|-------------|-------------|
|          | hours       | peak GB     | hours       | peak GB     | hours       | peak GB     |
| fp32     | 0.23        | 2.70        | 0.75        | 11.10       | 3.40        | 57.09       |
| bf16     | <b>0.07</b> | 1.33        | <b>0.12</b> | 5.50        | 0.39        | 28.57       |
| int8_wo  | 0.08        | 0.73        | 0.15        | 2.87        | 0.49        | 14.53       |
| fp8_dyn  | 0.13        | 0.75        | 0.17        | 2.90        | <b>0.38</b> | 14.57       |
| int8_dyn | 0.49        | 0.75        | 0.67        | 2.94        | 1.61        | 14.90       |
| int4_wo  | 0.39        | <b>0.60</b> | 1.23        | <b>1.82</b> | 5.47        | <b>7.91</b> |

## 5 Discussion

### 5.1 The two workloads want opposite configurations

This is the principal practical finding.

**Bulk embedding extraction** is compute-bound: large batches, large GEMMs. Quantization pays. fp8\_dyn with max-autotune wins on all three axes — 1.15× bf16 throughput, weights 5.29 → 2.65 GB, peak 12.63 → 7.24 GB, with accuracy indistinguishable from bf16.

**DMS variant-effect scoring** is latency-bound: the masked batch is small, no GEMM is large enough for low precision to help, and per-linear quantize/dequantize overhead is pure loss. bf16 in

| Measurement                             | Time               |
|-----------------------------------------|--------------------|
| eager, one forward at shape (11, 42)    | 27.9 ms            |
| eager, full masked_marginals pass       | 0.03 s             |
| compiled, one forward at shape (11, 42) | <b>15,138.1 ms</b> |

| Config   | Weights (GB) | Peak (GB) | Implied overhead                  |
|----------|--------------|-----------|-----------------------------------|
| fp32     | 2.49         | 2.56      | +0.07                             |
| bf16     | 1.21         | 3.74      | +2.53 $\approx$ fp32’s weights    |
| int8_wo  | 0.61         | 1.86      | +1.25 $\approx$ bf16’s weights    |
| int8_dyn | 0.61         | 1.26      | +0.65 $\approx$ int8_wo’s weights |

eager mode is fastest at every problem size, and `max-autotune` is actively harmful.

Applying the embedding-optimized configuration to DMS scoring makes it up to an order of magnitude slower. Because both workloads commonly appear in the same pipeline, this is a live risk rather than a theoretical one.

**This conflict is bounded by scale, and one model could not have shown that.** `fp8_dyn`’s per-call quantization overhead is fixed while useful work grows with hidden size, so the penalty shrinks as the model grows:  $0.30\times$  bf16 DMS speed at 650M,  $0.44\times$  at 3B,  $0.97\times$  at 15B (Table 14). By 15B, `fp8_dyn` is the right choice for *both* workloads at half the memory. The lesson is not that DMS is latency-bound — it is that whether a workload is latency-bound is a property of the model and the workload together.

## 5.2 On the interpretation of fidelity metrics

The conventional validation practice — quantize, measure drift against the full-precision model, accept if drift is small — is sound as a *risk bound*. Our data support the bound at scale: over 3015 assay/configuration pairs, fidelity loss correlates with the magnitude of ground-truth change at  $r = 0.74, 0.81$  and  $0.56$  at 650M, 3B and 15B, consistent with the  $0.87$  first estimated from 15 pairs.

It is not sound as a *quality ranking*. A configuration with larger drift is not thereby less accurate against reality; it is merely less predictable. The signed correlation is  $+0.10, -0.58$  and  $-0.40$  at 650M, 3B and 15B — it does not hold its sign across model scales.

A tempting misreading of our BLAT result is that quantization acts as beneficial regularization. We do not endorse this, and the full benchmark settles it: `int4_wo` is significantly *negative* at 650M, significantly *positive* at 3B, and null at 15B. Two scales would have permitted the reading that sensitivity falls with model size; the third removes it. A configuration cannot be genuinely more accurate than the reference from which it is derived in any way one could rely on in advance.

## 5.3 Averages conceal the failure that matters

The benchmark-mean result is that no configuration changes mean ground-truth correlation by more than  $0.007$ . Read alone, that licenses any configuration on the list. It should not.

`int8_dyn` at 3B is statistically indistinguishable from fp32 on the mean ( $p = 0.34$ ) and takes one assay from  $\rho = 0.591$  to  $0.223$ . A practitioner does not deploy against the mean of 201 assays; they deploy against one protein. The benchmark mean answers whether a configuration is safe *on average over targets*, which is a claim about a population, while the operative question is whether it is safe on a specific target.

The practical consequence is that the tail statistics — worst-case single-assay drift, and the count of assays exceeding a tolerance — are the numbers to select on, and they separate the configurations far more sharply than the means do. On that criterion `bf16` and `int8_wo` are safe (no assay beyond  $0.018$  across 402 assay/model pairs), `fp8_dyn` is nearly so ( $0.034$ ), and `int8_dyn` and `int4_wo` are not.

## 5.4 Recommendations

| Workload                  | Scale      | Recommendation                                                                                   |
|---------------------------|------------|--------------------------------------------------------------------------------------------------|
| Bulk embedding extraction | any (H200) | <code>fp8_dyn + max-autotune-no-cudagraphs</code>                                                |
| Bulk embedding extraction | any (A100) | <code>int8_dyn asymmetric</code> ; the symmetric default is unsafe (Table 15)                    |
| DMS / variant effect      | $\leq 3B$  | <code>bf16</code> , eager, no compile                                                            |
| DMS / variant effect      | 15B        | <code>fp8_dyn</code> — matches <code>bf16</code> speed at half the memory                        |
| DMS under memory pressure | $\leq 3B$  | <code>int8_wo</code> ( $\sim 70\%$ of <code>bf16</code> speed)                                   |
| Minimum memory footprint  | any        | <code>int4_wo</code> , only after confirming drift on <i>your</i> target; worst observed $-0.20$ |
| QLoRA fine-tuning base    | any        | <code>int4_wo</code>                                                                             |
| V100 / sm_70 hardware     | any        | Quantization buys memory only, never speed                                                       |

Two non-quantization measures outweigh most of the above. Moving from `fp32` to `bf16` is worth  $8.4\times$  on bulk extraction and  $6.2\times$  over the full DMS benchmark; length-bucketed batching eliminates 25% of wasted computation. Both should be applied before any quantization is considered.

A third, proposed in an earlier draft of this report, does not survive the full benchmark: the 650M checkpoint beat 3B by 0.14 Spearman on BLAT, which suggested model selection dominated precision selection. Over 201 assays the two are indistinguishable (0.4459 vs 0.4398, 95% CI  $[-0.0061, +0.0185]$ ,  $p = 0.35$ , with 650M ahead on 110 of 201). What remains true is narrower and still useful: the *uncertainty* on model choice ( $\pm 0.012$ ) exceeds every quantization effect measured here ( $\leq 0.007$ ). Precision is the settled variable; checkpoint choice is not, and is worth benchmarking on a representative assay.

## 6 Limitations

- **We still cannot predict which assays behave which way.** 201 assays establish the population behaviour of each configuration and the size of its tail. They do not identify in advance which target will be the one that collapses. The only reliable procedure remains scoring a candidate configuration against `fp32` on the actual target.
- **The 16 longest assays are excluded.** Assays beyond 1022 residues exceed ESM-2’s context and were skipped rather than truncated, since a truncation window is a scoring-protocol change that would confound the comparison. Configuration behaviour above  $L = 934$  is therefore unmeasured.
- **Multiple comparisons are uncorrected.** Five configurations are tested against `fp32` per model. Both significant results survive a Bonferroni threshold of  $0.05/5 = 0.01$ , but `int4_wo` at 650M sits exactly on it ( $p = 0.010$ ).
- **Bootstrap resolution.** At 2000 resamples the smallest reportable two-sided  $p$  is 0.001; values at that floor should be read as  $p < 0.002$  rather than as point estimates.
- **Low-signal assays resolve nothing individually.** Where  $\rho_{\text{expt}} \approx 0.3$  the model’s own error dominates and configuration differences fall below the noise floor; such assays contribute to the benchmark mean but carry little information alone.

- **Sequence-length coverage.** Bulk-throughput measurements used sequences of 512–1024 residues. Beyond  $L \approx 2000$  the  $O(L^2)$  attention term dominates peak memory and quantization’s share of the memory benefit shrinks correspondingly. A representative FASTA from the target workload would settle this.
- **Single random seed** for quantization; no ensemble over calibration variation.
- **The QLoRA track is unbuilt.** `int4_wo` at 1.61 GB is the natural base, and fine-tuning is the one setting where quantization buys capability rather than efficiency — full 3B fine-tuning requires roughly 45 GB of optimizer state alone.

## 7 Conclusions

**What to run.** Two workloads, and the answer depends on model scale.

- **Bulk embedding extraction: `fp8_dyn` with `max-autotune`.** Best on all three axes at once at 3B —  $1.15\times$  `bf16` throughput, peak memory  $12.63 \rightarrow 7.24$  GB, accuracy indistinguishable from `bf16`. Needs `sm_90`; on A100, `int8_wo`.
- **Variant-effect scoring up to 3B: `bf16`, `eager`, `uncompiled`.** Fastest at every problem size from  $L = 37$  to 934,  $6.2\times$  faster than `fp32` over the full benchmark. Quantization is pure overhead and compilation is actively harmful.
- **Variant-effect scoring at 15B: `fp8_dyn`.** Matches `bf16` speed ( $0.97\times$ ) at half the peak memory (14.6 vs 28.6 GB) with no measurable accuracy cost. The recommendation inverts with scale.
- **Avoid `int8_dyn` and `int4_wo` for variant-effect work at any scale.** Slower than the best option in this regime *and* owners of the two worst tails measured,  $-0.37$  and  $-0.20$ .

**What the evidence says about evidence.**

1. **Select on worst-case drift, not on benchmark means.** At no scale does any configuration move mean correlation by more than 0.007 — the statistic most likely to be quoted, and the one least able to distinguish these configurations. `int8_dyn` passes the mean test at 3B ( $p = 0.34$ ) and takes one assay from  $\rho = 0.591$  to 0.223. On means, `int8_dyn` and `bf16` differ by 0.002; on worst case, by 0.36 — a factor of 180.
2. **Drift from `fp32` is a valid safety certificate and an invalid quality ranking.** Over 3015 assay/configuration pairs it predicts the magnitude of ground-truth change ( $r = 0.56$ – $0.81$ ) and not its direction, whose correlation changes sign across model scales. `bf16` and `int8_wo` never exceed 0.018 Spearman on any of 402 assay/model pairs up to 3B and are safe on that basis alone, which is cheap to verify on a target with no experimental labels.
3. **Conclusions are bounded by model scale, not only by sample size.** The workload conflict holds at 650M and 3B and dissolves at 15B. The `int4_wo` effect is significantly negative, significantly positive, then null in ascending scale order — not a trend, so nothing to extrapolate. Even the MSA-depth stratification replicates at two scales and breaks at the third.

4. **Evidence scale is not a matter of rigour but of conclusions.** This study reached four successive verdicts. A synthetic mutational scan ranked configurations confidently and wrongly. Three real assays overturned it and produced two further generalisations — that quantization does not systematically degrade accuracy, and that checkpoint choice dominates precision choice — both reversed by the full 201-assay benchmark. That benchmark at two scales then suggested sensitivity grows with model size, which the third scale removed. Every intermediate conclusion was statistically sound on its own data. The full three-scale benchmark costs 16.2 GPU-hours.

**The general claim.** A benchmark mean answers whether a configuration is safe *on average over targets*. Deployment asks whether it is safe on *one* target. Here those questions differ by the gap between 0.002 and 0.36. Wherever a benchmark aggregates over heterogeneous instances and the user faces one instance, we would report the tail alongside the mean as a matter of course.

## 8 Reproduction

```
# Environment (handles glibc/GCC/cache-quota constraints)
source env/activate.sh

# Fetch assay data; verifies variant indexing against target_seq
python src/fetch_proteingym.py IF1_ECOLI_Kelsic_2016 \
    BLAT_ECOLX_Stiffler_2015 HSP82_YEAST_Flynn_2019

# Bulk throughput + fidelity matrix (compiled)
sbatch -p H8V141_SAP112M2000_L --mem=200G slurm/run_matrix.sbatch \
    3B fp32,bf16,int8_wo,int8_dyn,fp8_dyn,int4_wo --compile

# Real-assay DMS: accuracy, speed, memory (uncompiled by design)
sbatch -p H8V141_SAP112M2000_L --mem=200G slurm/run_dms.sbatch \
    3B fp32,bf16,int8_wo,int8_dyn,fp8_dyn,int4_wo

# Significance of ground-truth shifts (single assay, variant bootstrap)
python src/analyze_dms.py results/dms_3B*_<jobid>_scores.json

# --- Full benchmark -----
# All 217 assays; validates every variant's WT and refuses to write on
# mismatch. ~110 MB of parquet, cached in data/proteingym_v1/_raw
python src/fetch_proteingym_v1.py

# One array task per config; each loads its model once and scores all
# 201 assays. Resumable: a resubmission skips assays already written.
sbatch --array=0-5 -p H8V141_SAP112M2000_L --mem=200G \
    slurm/run_proteingym.sbatch 3B
sbatch --array=0-5 -p H8V141_SAP112M2000_L --mem=200G \
    slurm/run_proteingym.sbatch 650M
# 15B needs more memory and wall time (int4_wo alone is 5.5 h)
sbatch --array=0-5 -p H8V141_SAP112M2000_L --mem=300G --time=12:00:00 \
    slurm/run_proteingym.sbatch 15B

# The two W8A8 repairs: asymmetric activations, and SmoothQuant with
# 16 calibration sequences.
sbatch --array=0-1 -p H8V141_SAP112M2000_L --mem=200G \
    slurm/run_proteingym.sbatch 3B int8_dyn_asym,int8_dyn_sq

# Benchmark-wide accuracy / speed / RAM + protein-clustered bootstrap
```

```

python src/aggregate_proteingym.py --model 3B
python src/aggregate_proteingym.py --model 650M
python src/aggregate_proteingym.py --model 15B

# --- Analyses reported here -----
# Pareto surface, interaction contrasts, conditioned tail, aggregation
# conventions with post-stratification, and the SNR screen
python src/analyze_scales.py

# W8A8 repair, per assay -> results/proteingym/w8a8_remedies.csv
python src/analyze_w8a8.py

# Repeatability floor: rep2 against the primary run, bitwise
python src/analyze_repeat.py

# Per-channel activation profile of the collapsing assay vs controls
sbatch -p H8V141_SAP112M2000_L --mem=200G slurm/run_outliers.sbatch

# Figures (also used by the preprint)
python src/figures.py

```

Result artifacts referenced in this report:

- Throughput matrices (SLURM jobs 5393462, 5393520, 5393524, 5393554):  
results/matrix.3B-<jobid>.json
- ProteinGym, 3B, with per-variant scores (job 5394386):  
results/dms\_3B-<assay>\_5394386.json
- ProteinGym, 650M (job 5394316): results/dms\_650M-<assay>\_5394316.json
- Full benchmark, per-variant scores: 3B (array 5395036), 650M (array 5395037), 15B (array 5395122):  
results/proteingym/<model>\_<config>.jsonl.gz
- Full-benchmark analysis: results/proteingym/summary-<model>.json and  
summary-<model>\_per\_assay.csv (1206 rows: one per assay/configuration)